Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	28-10- 2026
Team ID	EN25CS3010483
Project Name	EV Market Performance and Insights
Maximum Marks	4 Marks

Functional Requirements:
Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Connect Tableau to the EV market dataset (CSV / Excel) covering vehicle sales, price, battery specifications, driving range, manufacturer, manufacturing country, customer ratings, market region and CO2 metrics; relate or join the tables on common fields.	High
2	Data Processing	Clean and standardize the data (data types, null values, duplicates, category names); assign geographic roles to country and region; prepare fields such as battery type, drivetrain, vehicle category and model year for analysis.	High
3	Business Analysis	Calculate the KPIs and measures: total EV sales, average vehicle price, average vehicle range, CO2 saved (tons per year), 3-year retained value, customer rating, battery cycle life, maximum DC charging power and energy efficiency (Wh per km); aggregate them by category, manufacturer, battery type, country, region and year.	High
4	Business Insights & Analytical Outputs	Provide two interactive dashboards. Dashboard 1: KPI tiles, production output by country and category, top manufacturers by customer review and sales units, driving range distribution, and 3-year retained value. Dashboard 2: sales trend over time, battery technology performance, charging vs efficiency, and regional CO2 saved by drivetrain.	High
5	User Interaction & Analysis	Let users filter by vehicle category through the shared legend, highlight and sort data, use tooltips, and compare categories, manufacturers, battery technologies, countries and regions across the charts.	Medium

6	Data Update & Reporting	Refresh the data source so that all worksheets and dashboards update, and make the dashboards shareable and exportable (image / PDF) for executive and stakeholder reporting.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Dashboards use clear titles, a consistent color legend by vehicle category, tooltips and interactive filters so executives and analysts can understand insights without training.
NFR-2	**Security**	Access to the dataset and workbook is limited to authorized users, and dashboards are shared with view-only permission. The data covers market-level information and contains no personal customer data.
NFR-3	**Reliability**	Dashboards show accurate and consistent values after every refresh; calculated fields are validated against the source data and data connection errors are reported clearly.
NFR-4	**Performance**	Dashboards load and respond to filters within a few seconds by using extracts, aggregated measures and a limited number of marks per chart.
NFR-5	**Availability**	Published dashboards are accessible online to stakeholders whenever needed, and a packaged workbook copy is kept as a backup.
NFR-6	**Scalability**	The design supports more rows, new model years, manufacturers and vehicle categories by refreshing the data extract, without redesigning the dashboards.